

What's the Best Ski Lift in North America?

An essential property of every ski lift is that the average slope of the terrain it serves cannot be greater than the slope of the lift itself. This is, of course, an obvious corollary of the Fundamental Theorem of Calculus. But it is intuitive to any skier. You can meander down a cruiser gentler than the lift line, but if you want steeper terrain than the lift itself, you'll need to traverse for the spicy bits—or accept a long, flat runout home.

That makes the average slope of a lift line—its rise over run—a surprisingly powerful summary statistic for both the terrain it serves and the awesomeness of the lift itself. Awesomeness, of course, is multidimensional: terrain quality, line speed, lap-worthy vert, the odds that Jerry blows your line, and the visual exposure to the après crowd below. But skiing has a proud tradition of collapsing subjective greatness into a single number (see e.g. Gaffney, 2003, on “radness”). With a little geometry, we can build a novel and universal measure of what a lift actually delivers.

To fix ideas, take KT-22, perhaps North America's most storied lift, raising 2,100 skiers an hour up 1,755 vertical feet. It's iconic in part because it serves sandy terrain, and rockets up straight from the base, allowing your legions of prospective fans to ooh and aww at your GNAR.

A central element of KT-22's value proposition is that it is steep. It raises you 1,755 feet over a 5,042 rope length, which implies, thanks to the Pythagorean theorem, that its average gradient is about 37 percent. That is about the floor on a solid black diamond. If you wanted to lap black-diamond terrain all day, just follow the lift line.

After crunching the data for every lift in North America, a pattern emerges: steep is awesome. There are outliers—and they prove the rule.

The very steepest lifts are odd ducks. Seventh Heaven at Stevens Pass hits 66 percent—the steepest fixed-grip chairlift in North America—but at just 437 vertical feet on a 1960 Riblet double, it's more of a steep sprint than a proper lap. The platter at Nakiska clocks in at 64 percent, built to serve the 1988 Olympic downhill and never opened to the public. The Lone Peak Tram at Big Sky hits 60 percent and delivers not single-black, not double-black, but triple-black terrain. Patrol ropes off the truly absurd pitches, leaving a tangle of wire more tempting to clip into than duck under. Snowbasin's Allen Peak Tram (48 percent) was constructed for the 2002 men's downhill, dropping racers into a 74-degree start.

Among regular, publicly accessible lifts, Palisades Tahoe stands out because it's stacked with steep chairs. Beyond KT-22, the Scott Triple clocks in at 54 percent—the steepest lift in the dataset serving true skiable terrain—and the runs beneath it are appropriately double black. Add Red Dog (43 percent), Shirley Lake (47 percent), and the rest, and Palisades' five steepest lifts average 45 percent—making it the steepest major lift system in North America.

Who's next?

- **Telluride** ranks second. Revelation (48 percent) and Gold Hill (44 percent) anchor a top-five average of 43 percent.
- **Whiteface** takes third—East Coast represent—led by the 48-percent Summit Quad and a matching 43-percent top-five average.
- **Jackson Hole** comes in fourth. Teton (47 percent), Thunder (44 percent), Sublette (43 percent), and the tram combine for a 42-percent top-five average.
- **Stevens Pass** lands fifth. Double Diamond (48 percent) and Tye Mill (53 percent) power a 42-percent average across its five steepest lifts.
- **Snowbird** sits sixth, where Mineral Basin (45 percent), Little Cloud (44 percent), and the Peak Tram push the top-five average to 41 percent.

And seventh? **Montana Snowbowl**—four lifts, no high-speed anything, and a system-wide weighted slope of 41 percent that outpaces every mega-resort in North America.

And there are legends among the individual lifts, too: Kachina Peak at Taos and the Forerunner Quad at Stowe—both tied with KT-22 in average grade—proving that greatness, like fall line, transcends geography.

The Steepest Individual Lifts

Lift	Resort	Slope	Vertical	Type
Honeycomb Return	Solitude	58%	655 ft	Quad
Scott	Palisades Tahoe	54%	1,068 ft	Triple
Kachina Peak	Taos	53%	1,095 ft	Triple
Tye Mill	Stevens Pass	53%	706 ft	Triple
Chair 5	Mt. Baker	53%	1,271 ft	Quad
Deep Temerity	Aspen Highlands	52%	1,708 ft	Triple
Eagle	Park City	52%	1,155 ft	Triple
Double Diamond	Stevens Pass	48%	2,466 ft	Triple
Revelation	Telluride	48%	799 ft	Quad
Chair 1	Taos	48%	1,633 ft	HSQ
Summit	Whiteface	48%	1,852 ft	Quad
Teton	Jackson Hole	47%	1,722 ft	HSQ
Shirley Lake	Palisades Tahoe	47%	1,109 ft	HS6

Gunbarrel Express	Heavenly	47%	1,588 ft	HSQ
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Field Notes on the Legends

Revelation (Telluride): "You unload at 12,500 feet into oxygen debt and existential reflection."

Summit Quad (Whiteface): "Almost as steep as your odds of frostbite."

Scott Chair (Palisades): "Drop a pole at the top; it'll be waiting for you at the base lodge."

Honeycomb Return (Solitude): "A misnomer: 58 percent up, but no fall-line back."

Lone Peak Tram (Big Sky): "Where the steepest ride is often the return trip in the tram."

What the Numbers Don't Tell You

Then there are the lifts that defy the premise. Lake Louise's Great Divide chair runs almost perfectly flat—a triumph of civil engineering over gravity. Whistler's Peak 2 Peak was engineered with a slope of zero, the world's most spectacular horizontal. The airport people mover has more vertical.

The Canyons—Park City gondola epitomizes the problem. Built to stitch two resorts into one, it climbs at just 14 percent—a mechanized sidewalk. You ride it once for novelty, then avoid it forever. Park City may now be the largest ski resort in America, but its 22 lifts average a weighted slope of just 26 percent. Breckenridge, with 18 lifts, manages 22 percent. Size isn't quality.

And then there's Alta—the paradox of steep skiing. It's legendary for its pitch, its powder, and its posture of moral superiority toward anyone on a snowboard. But according to the math, its lifts aren't especially steep. None crack the top hundred.

Which helps explain why Alta is also legendary for its traverses. The lift lines are mellow; the terrain is not. Stop in the wrong place and you'll hear about it. The mountain functions as a network of conveyor belts feeding an economy of lateral ambition—skiers shuffling and sidestepping in pursuit of vertical the lifts can't quite deliver.

In the adjacent canyon, Solitude manages the opposite problem. Its steepest lift—Honeycomb Return, at a hair-raising 58 percent—serves remarkably little skiable fall line. Meanwhile, much of the mountain's best terrain sits behind Summit, a lift that somehow manages to descend for stretches of its route. Completing a full Honeycomb lap requires Moonbeam → Apex → Summit → Honeycomb Return, a logistical obstacle course better suited to an industrial engineer than a powder hound. Solitude is a mountain of beautiful terrain connected by lifts that quietly insist: you can't get there from here.

There's something to be said for the opposite approach—the old-school New England model. Places like Stowe or Cannon, where the lift runs from base to summit and the skiing runs straight back down the fall line, like God and the CCC intended. No traverses. No mid-mountain

detours. No four-lift Rube Goldberg laps. The lift points uphill, the runs point down, and everyone's back at the car in time for a parking-lot beer. It's an elegance born of rock, weather, and Yankee pragmatism—and proof that sometimes the shortest path between two points is also the best ski run.

The Hidden Gems

The data doubles as a trip-planning tool. Sort by weighted slope, remove the resorts you've heard of, and you get a list of places worth visiting.

Montana Snowbowl. The seventh-ranked system from above, it sits outside Missoula with 2,600 vertical feet of fall-line skiing. Lift tickets are \$62.

Mt. Norquay. Twenty minutes from Banff, it quietly delivers a 40-percent weighted slope while tourists queue at Sunshine Village. Three lifts. North-facing terrain. Open since 1926.

Castle Mountain. Straddling the Alberta–BC border, it comes in at 37 percent, with thousands of acres of permitted terrain still undeveloped.

Snow King. The town hill in Jackson, it climbs 1,571 vertical feet straight from downtown and posts a 36-percent weighted slope—steeper than Snowbird.

Pajarito Mountain. In Los Alamos, it hits 34 percent, serving steep terrain to a clientele of physicists who don't discuss their day jobs on the chairlift.

Then there are the single-lift holdouts. Turner Mountain in northwest Montana runs 2,110 vertical feet on a volunteer-built double chair—no grid electricity, no cell service, \$45 tickets. Silverton Mountain operates one 1978 double chair for 80 skiers a day; the lift is merely transportation to the hiking.

None of these places have village plazas. What they have are steep lift lines and short queues. The math suggests they offer more skiing per dollar than resorts charging three times as much.

KT-22 moves 2,100 skiers per hour at a 37 percent grade: 777 “steep-percent-skiers per hour.” Honeycomb Return, by contrast, runs half that capacity at 58 percent, giving only 609 steep-percent-skiers per hour. In other words, KT delivers more black-diamond joy to more people, which feels morally correct.

In the end, steepness is only one dimension of lift quality—just as vertical is only one measure of joy. But it's a pretty good start. The math may explain why we line up for KT instead of taking the groomer gondola: gravity has an elegance that no marketing slogan can touch.

Analysis includes 1,500+ operating chairlifts in the US and Canada. Scenic gondolas, amusement park rides, and single-lift resorts (with exceptions for legitimate ski areas like Silverton and Turner Mountain) excluded. Minimum thresholds: 500 feet vertical, 5% slope, 500 skiers/hour capacity.

Slope = vertical rise / horizontal distance, where horizontal = $\sqrt{\text{length}^2 - \text{vertical}^2}$. Weighted slope = $\Sigma(\text{capacity} \times \text{vertical} \times \text{slope}) / \Sigma(\text{capacity} \times \text{vertical})$ across all lifts at a resort. Top-5 weighted slope uses the same formula applied to the five steepest lifts only.